

Fig5 prototype: strict-path synten/recombination schematic

Geometry comes from selected_segments.tsv; autosomal PHR rows are candidate exchange-compatible patches, not event-level validation.

Flow legend PAR1 positive-control flow primary PHR donor flow side fragment low-confidence tail same-chromosome context

1. PAR1 positive control

PAN027 paternal hap2 <- PAN011 father

PAR 151 kb; same-chr 349 kb
recombinant/child: PAN027#2#chrX.paternal:12265-512264_chrX_parm
chrX:12265-512265

parental same-chromosome context

chrXp

PAN011 haplotype1 chrXp chrX:3,251-503,251

non-homologous PAR1 source

chrYp

PAN011 haplotype2 chrYp chrY:5,839-505,839

Full chromosome/arm context; bands are neutral schematic bands from chromosome-size context, not exact cytobands.

Child/query native window: chrX:12,265-512,265

2. PHR candidate 1: PAN027 chr9q with chr3q donor

PAN027 paternal hap2 <- PAN011 father

primary donor 45 kb; side fragments 8 kb; low-conf. tail 493 bp; same-chr 446 kb
recombinant/child: PAN027#2#chr9.paternal:135704825-136204824_chr9_qarm
chr9:135704825-136204825

parental same-chromosome context

chr9q

PAN011 haplotype1 chr9q chr9:135,595,959-136,095,959

PAN011 haplotype2 chr9q chr9:133,240,004-133,740,004

primary non-homologous PHR donor

chr3q

PAN011 haplotype2 chr3q chr3:205,075,647-205,575,647

PAN011 haplotype1 chr3q chr3:202,049,714-202,549,714

secondary side fragments / caveats

chr15q, chr16q, chr20q

PAN011 haplotype2 chr16q chr16:89,999,497-90,499,497

PAN011 haplotype1 chr15q chr15:95,856,803-96,356,803

PAN011 haplotype1 chr20q chr20:66,002,904-66,502,904

Full chromosome/arm context; bands are neutral schematic bands from chromosome-size context, not exact cytobands.

Child/query native window: chr9:135,704,825-136,204,825

3. PHR candidate 2: PAN028 strict chr9q path

PAN028 maternal hap1 <- PAN027 mother

primary donor 34 kb; side fragments 16 kb; same-chr 449 kb
recombinant/child: PAN028#1#chr9.haplotype1:134380985-134880984_chr9_qarm
chr9:134380985-134880985

parental same-chromosome context

chr9q

PAN027 paternal chr9q chr9:135,704,825-136,204,825

PAN027 maternal chr9q chr9:128,564,684-129,064,684

primary non-homologous PHR donor

chr3q

PAN027 maternal chr3q chr3:200,905,557-201,405,557

PAN027 paternal chr3q chr3:202,048,279-202,548,279

secondary side fragments / caveats

chr15q, chr16q

PAN027 paternal chr16q chr16:89,997,300-90,497,300

PAN027 paternal chr15q chr15:98,308,884-98,808,884

Full chromosome/arm context; bands are neutral schematic bands from chromosome-size context, not exact cytobands.

Child/query native window: chr9:134,380,985-134,880,985